PUMP BEARING REPLACEMENT PROCEDURE

Document ID: PUMP-SOP-002

Revision: 1.0

Asset Class: Centrifugal Pump

Purpose: Safe Removal, Inspection and Replacement of Pump Bearings

---

1. Purpose

This procedure defines the standardized methodology for replacing pump bearings while minimizing equipment damage and ensuring safe operation.

The procedure applies when:

- Bearing Failure is confirmed
- Vibration exceeds acceptable limits
- Bearing temperature exceeds allowable limits
- Bearing wear is identified during inspection
- AI Copilot recommends bearing replacement

---

2. Safety Requirements

Before beginning work:

- Obtain maintenance permit
- Obtain equipment isolation approval
- Notify operations team
- Complete Job Safety Analysis (JSA)

---

3. Required PPE

Mandatory PPE:

- Safety Helmet
- Safety Glasses

- Gloves
- Safety Shoes
- Hearing Protection

Additional PPE:

- Chemical-resistant gloves if process contamination exists

---

4. Lockout Tagout (LOTO)

Step 1:

Stop pump.

Step 2:

Open motor isolation breaker.

Step 3:

Apply lock.

Step 4:

Apply tag.

Step 5:

Verify zero energy condition.

Step 6:

Drain and depressurize pump.

---

5. Removal Procedure

Coupling Removal

Inspect coupling.

Record condition.

Remove coupling guard.

Mark alignment position.

Disconnect coupling.

Bearing Housing Access

Remove:

- Bearing cover
- Lubrication lines
- Instrument connections

Inspect:

- Oil contamination
- Metal particles
- Overheating signs

Bearing Removal

Use approved puller.

Never use hammer impact directly on shaft.

Inspect:

- Inner race
- Outer race
- Rolling elements
- Cage assembly

---

6. Bearing Inspection Criteria

Reject Bearing If:

- Pitting exists
- Spalling exists
- Cracks exist
- Excessive wear exists
- Heat discoloration exists

---

7. Root Cause Inspection

Determine failure origin.

Investigate:

- Lubrication failure
- Misalignment
- Overload
- Contamination
- Cavitation-induced vibration

---

8. New Bearing Installation

Verify:

- Correct bearing type
- Correct part number
- Correct lubricant

Install using approved tools.

Apply force only through appropriate race.

Never transmit installation force through rolling elements.

---

9. Alignment Verification

Perform:

- Laser alignment

or

- Dial indicator alignment

Alignment must comply with refinery standards.

---

10. Post Maintenance Verification

Verify:

Bearing Temperature

Motor Current

Vibration

Flow Rate

Discharge Pressure

---

11. Acceptance Criteria

Bearing Temperature:

<75°C

Vibration:

<4 mm/s

Motor Current:

Within normal operating range

No abnormal noise.

12. AI Copilot Validation Rules

Maintenance considered successful if:

Bearing Temperature decreases.

Vibration decreases.

Motor Current stabilizes.

Failure Probability decreases.

Predicted RUL increases.

13. Lessons Learned Recording

Record:

- Root Cause
- Corrective Action
- Telemetry Before Repair
- Telemetry After Repair
- Downtime
- Maintenance Cost